Decorators in Python -

what is decoration -

adding things to somethings to make more attractive / presentable.

- what is decorator in Python

Decorator function which takes other function as input, add additional functionalities & returns it

① Suppose we have function. & we have to modify this function. we want to add additional functionalities to it. we will provide this function as a input

to the decorator function.

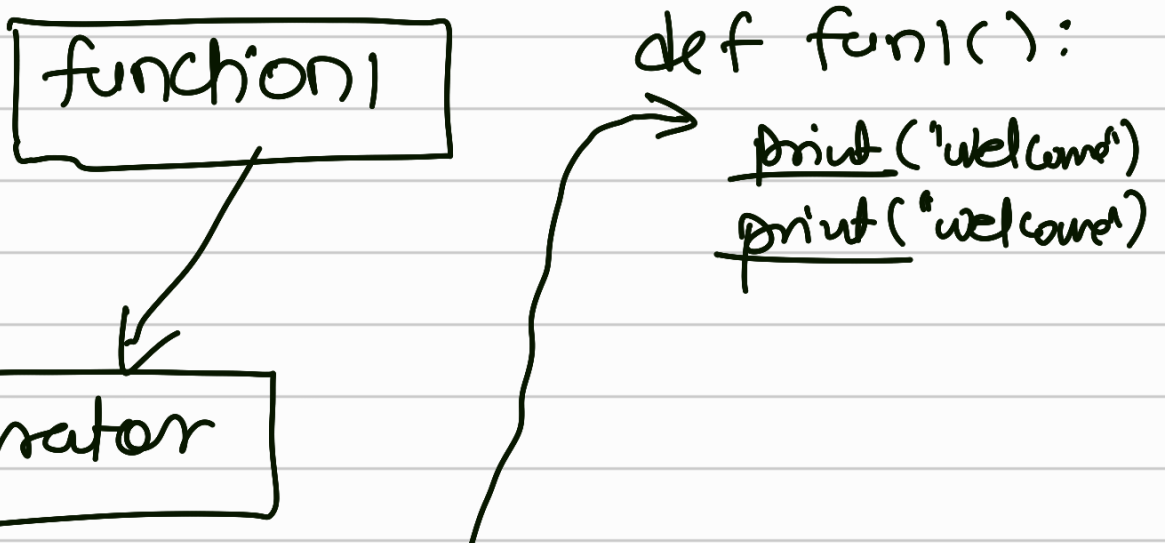
2) Decorator is also a function which takes this function (which we want to modify) as input, will perform modification & returns modified function




```
def fun1():
    print("welcome")
    print("welcome")

fun1()
```

```
def dec(fun):
    fun()
    print("welcome")
    print("welcome")
```



```
print("welcome")
```
← welcome

addition

```
def decor(func):
    def inner():
        result1 = func()          # 30
        num3 = int(input("Enter third no:"))   # 40
        final_result = result1 + num3          # 30+40=70
        return final_result                    # → 70
    return inner

def addition():
    num1 = int(input("Enter first NO:"))       # 10
    num2 = int(input("Enter second NO:"))      # 20
    result = num1 + num                        # 10+20=30
    return result

addition = decor(addition)      # inner
print("addition is", addition())
```

70

O/p - addition is 70